CBC HomeLearning Platform

Tech Stack: MVP (Free) → Production (Scalable)

March 2026 | Engineering Reference

HOW TO USE THIS DOCUMENTEvery section follows the same pattern: start with the FREE tier to build and validate the MVP, then upgrade each component only when a specific trigger is hit (user volume, data size, performance degradation, or revenue milestone). You never need to migrate everything at once — upgrade one layer at a time.

1. The Free-First Strategy

The approach is simple: use genuinely free tools — not free trials — to build and launch the MVP with zero or near-zero infrastructure cost. This lets you validate the product with real Kenyan homeschooling families before spending money. Every component below has a clear upgrade path triggered by a specific condition, not an arbitrary date.

Three important principles for this strategy:

- Free means genuinely free, not 14-day trials. Every MVP tool listed below has a permanent free tier or is fully open-source with no usage limits that will kill you before you have paying users.
- Upgrade triggers are specific. You should only upgrade when a defined threshold is hit — not "when things feel slow". This prevents premature spending.
- The stack is designed so each layer can be upgraded independently. Swapping your database does not require rewriting your frontend. Swapping your LLM does not require changing your RAG pipeline.

ESTIMATED MVP INFRASTRUCTURE COST: KES 0 – 3,000/month (approximately $0–$23 USD). The only likely early cost is the Anthropic API for LLM calls, which you can control tightly with caching and rate limits.

2. Frontend — Web & Mobile

Layer / Component	MVP — Free Tier	Production — Scalable/Paid
Framework	**Next.js 14 (free, open-source)** Deploy on Vercel free tier (Hobby). Supports up to 100GB bandwidth/month.	**Next.js 14 (same — no change needed)** Upgrade to Vercel Pro ($20/mo) only when you exceed free bandwidth or need team collaboration features.
Language	**TypeScript (free, always)** No cost at any scale. Use strict mode from Day 1 to avoid painful refactors later.	**TypeScript (same — no change needed)** No upgrade needed. TypeScript scales with you for free.
UI Components	**shadcn/ui + Tailwind CSS (free, open-source)** Install shadcn/ui components locally — they live in your codebase, not a paid subscription.	**shadcn/ui + Tailwind CSS (same)** No upgrade needed. Add a paid design system (e.g. Tremor Pro) only if you need advanced data viz.
State Management	**Zustand + TanStack Query (free, open-source)** Both are permanently free open-source libraries. No usage limits.	**Zustand + TanStack Query (same)** No upgrade needed at any scale.
Mobile App	**Expo (React Native) — free tier** Expo Go for development and testing is completely free. Free tier allows limited cloud builds.	**Expo EAS Build ($99/mo) or self-hosted builds** Upgrade to Expo EAS when you need unlimited production builds and OTA updates for paying users.
Offline / PWA	**next-pwa + Workbox (free, open-source)** Service worker caching is free and critical for Kenyan low-bandwidth users.	**Same + Cloudflare Workers for edge caching** Add Cloudflare Workers ($5/mo) for smarter edge caching when user base grows nationally.
Charts & Visualisations	**Recharts (free, open-source)** Recharts handles all MVP charts: mastery heatmaps, progress lines, competency radar.	**Recharts + Observable Plot for complex views** Add Observable Plot for advanced longitudinal analytics when you have 500+ learners with rich data.

3. Backend — API & Business Logic

Layer / Component	MVP — Free Tier	Production — Scalable/Paid
API Framework	**Next.js API Routes (free)** All API logic in Next.js API routes initially. No separate server needed.	**Next.js API Routes + FastAPI (Railway paid)** Extract heavy AI/ML processing (RAG, voice, timetable) to FastAPI microservices on Railway when API routes become slow (~$5–20/mo per service).
Authentication	**NextAuth.js v5 / Auth.js (free, open-source)** Supports email magic links, Google OAuth, credentials. Stores sessions in Supabase free tier.	**NextAuth.js v5 (same — no change needed)** No paid upgrade needed. Add Clerk ($25/mo) only if you need enterprise SSO for school partnerships.
Real-time Updates	**Supabase Realtime (free tier: 200 concurrent connections)** Free tier supports up to 200 concurrent WebSocket connections — enough for MVP beta.	**Supabase Pro ($25/mo — 500 concurrent)** Upgrade to Supabase Pro when you exceed 200 concurrent users or 500MB database size.
Background Jobs	**Vercel Cron Jobs (free tier: 2 cron jobs) + in-process queues** Use Vercel Cron for scheduled tasks (SBA reports, weekly summaries). In-process for simple async jobs.	**Upstash QStash ($10/mo) or Railway + BullMQ** Upgrade to BullMQ + Redis when you need reliable job queuing with retry logic for lesson generation at scale.
Caching	**In-memory caching (Node.js Map) + Supabase query cache** Cache generated lessons in memory during request lifecycle. Use Supabase for persistent caching.	**Upstash Redis (Pay-per-use, ~$10/mo)** Add Upstash Redis when RAG generation costs become significant and lesson cache hit rate needs tracking.
API Type Safety	**tRPC (free, open-source)** Type-safe API layer between Next.js frontend and backend. Zero runtime cost.	**tRPC (same — no change needed)** No upgrade needed at any scale.

4. AI & ML Stack

This is the section where costs appear earliest, because LLM API calls cost money. The strategy here is to minimise API calls through aggressive caching of generated lessons, and to use free/local models wherever possible for voice processing.

Layer / Component	MVP — Free Tier	Production — Scalable/Paid
Primary LLM	**Claude Haiku via Anthropic API (cheapest Claude model)** Use claude-haiku-4-5 for lesson generation during MVP. ~$0.25 per million input tokens. Cache every generated lesson — don't regenerate unless curriculum changes.	**Claude Sonnet 4 via Anthropic API** Upgrade to claude-sonnet-4-20250514 when lesson quality feedback from families indicates Haiku is insufficient, or when you have revenue to absorb the cost (~3–4x price of Haiku).
LLM Orchestration	**LangChain (free, open-source)** LangChain for simple RAG chains in MVP. No cost — pure Python library.	**LangGraph (free, open-source) — multi-agent** Add LangGraph for the full multi-agent architecture (12 agents) when Phase 2 begins. Still free/open-source, just more complex.
Vector Database	**Weaviate Free Cloud (1 sandbox cluster, 1GB)** Weaviate's free sandbox is sufficient for Grade 4 English MVP (~150 lessons, ~5,000 chunks).	**Weaviate Cloud Starter ($25/mo) or self-hosted on Railway** Upgrade when you ingest multiple grades/subjects and sandbox storage fills. Self-host on Railway (~$10/mo) as a cost-effective middle step.
Embeddings	**sentence-transformers / multilingual-e5-large (free, local)** Run locally via sentence-transformers Python library. Free at any volume. Requires ~500MB RAM.	**Same — or OpenAI text-embedding-3-small for speed** Switch to OpenAI embeddings ($0.02/million tokens) only if local embedding becomes a latency bottleneck at high ingestion volumes.
Knowledge Graph	**Neo4j Community Edition (free, self-hosted)** Self-host Neo4j Community on Railway free tier (512MB RAM) for MVP curriculum graph.	**Neo4j Aura Professional ($65/mo) or self-hosted on Railway** Upgrade to Aura Professional when graph size exceeds free tier limits or you need HA/backups for production.
STT (Voice Input)	**Whisper.cpp (free, local) + Web Speech API (free, browser)** Run whisper.cpp locally on your server — completely free, no per-call cost. Web Speech API as browser fallback.	**OpenAI Whisper API ($0.006/min) for cloud fallback** Add OpenAI Whisper API as a fallback for users with low-powered devices that can't run local inference.
TTS (Voice Output)	**Kokoro-82M (free, open-source, local)** Kokoro runs on CPU, completely free, good quality. Cache common phrases (lesson intros, instructions) so TTS is rarely called.	**ElevenLabs API ($5/mo starter — 30,000 chars)** Add ElevenLabs for premium voice quality in paid subscription tiers. Keep Kokoro as free-tier voice.
Knowledge Tracing	**PyBKT (free, open-source Python)** PyBKT is a well-validated Bayesian Knowledge Tracing library. Free. Sufficient for MVP mastery tracking.	**Same + custom neural KT model** Build a custom neural KT model (Deep Knowledge Tracing) when you have 1,000+ learner-interaction records to train on.

5. Database & Storage

Layer / Component	MVP — Free Tier	Production — Scalable/Paid
Primary Database	**Supabase Free Tier (500MB, 2 projects)** Free tier is generous: 500MB PostgreSQL, built-in auth, row-level security, realtime. Perfect for MVP with 20–50 families.	**Supabase Pro ($25/mo — 8GB, daily backups)** Upgrade to Pro when database approaches 400MB, or when you need daily automated backups for SBA/compliance records.
ORM	**Prisma (free, open-source)** Prisma ORM is permanently free. Use it from Day 1 for type-safe queries and migration management.	**Prisma (same) + Prisma Accelerate ($10/mo) for connection pooling** Add Prisma Accelerate for connection pooling when you have many concurrent users hitting the database.
Portfolio Storage	**Supabase Storage (free tier: 1GB)** Store practical session photos and audio in Supabase Storage free tier (1GB — enough for MVP beta).	**Cloudflare R2 ($0.015/GB after 10GB free)** Migrate to Cloudflare R2 when storage exceeds 1GB. R2 has zero egress fees — critical for Kenyan bandwidth cost.
Cache / Session Store	**Supabase + in-memory (free)** Use Supabase for persistent session data. Node.js in-memory Map for hot cache during MVP.	**Upstash Redis ($10/mo pay-per-use)** Add Upstash Redis when you need distributed caching across multiple server instances.
Full-Text Search	**Supabase pg_tsvector (built-in, free)** PostgreSQL full-text search (pg_tsvector) is built into Supabase — free and surprisingly powerful for MVP.	**Meilisearch Cloud ($30/mo) or self-hosted Typesense** Upgrade to Meilisearch when search latency becomes noticeable (>500ms) or you need typo-tolerance for lesson/content search.

6. Infrastructure & DevOps

Layer / Component	MVP — Free Tier	Production — Scalable/Paid
Frontend Hosting	**Vercel Hobby (free — 100GB bandwidth)** Vercel Hobby is free forever for personal/small projects. Handles Next.js perfectly. 100GB bandwidth/month is enough for MVP.	**Vercel Pro ($20/mo — unlimited bandwidth)** Upgrade to Pro when you have a team (need collaboration) or exceed 100GB bandwidth with active users.
Backend / AI Services	**Railway Starter ($5/mo credit, ~free for small services)** Railway gives $5 of free credit monthly. A small FastAPI service for RAG easily fits within this. Excellent DX.	**Railway Team ($20/mo) or AWS ECS** Upgrade to Railway Team when you need more compute for AI services, or migrate to AWS ECS for enterprise scale.
CI/CD	**GitHub Actions (free — 2,000 min/month for private repos)** Free tier is more than enough for MVP: automated tests, type checks, deployments on every PR.	**GitHub Actions (same — upgrade to Team plan if needed)** Upgrade GitHub plan only if you exceed 2,000 CI minutes/month (unlikely until large team).
Error Monitoring	**Sentry Free (5,000 errors/month)** Sentry free tier catches all errors during MVP. Set up on Day 1 — don't fly blind.	**Sentry Team ($26/mo — 50,000 errors)** Upgrade when error volume exceeds free tier, or when you need performance monitoring (transaction tracing).
Product Analytics	**PostHog Cloud Free (1M events/month)** PostHog free tier allows 1 million events per month — enormous for an MVP. Privacy-first, GDPR compliant.	**PostHog Cloud Paid ($0/mo up to 1M, then usage-based)** PostHog scales gracefully. You likely won't hit the free tier until you have thousands of daily active users.
CDN / Security	**Cloudflare Free (CDN, DDoS, WAF basics)** Cloudflare free tier provides full CDN, DDoS protection, and basic WAF. Has a Kenya edge PoP. Use from Day 1.	**Cloudflare Pro ($20/mo — advanced WAF, analytics)** Upgrade to Pro for advanced security rules and better analytics when you have paying users to protect.
Email (Transactional)	**Resend Free (3,000 emails/month)** Resend free tier handles all parent notifications, SBA reports, and alerts for MVP (3,000 emails = ~150 families getting daily emails).	**Resend Pro ($20/mo — 50,000 emails)** Upgrade to Pro when you exceed 3,000 emails/month — around 150–200 active families.
Push Notifications	**Firebase Cloud Messaging (free, always)** FCM is free at any scale. No upgrade path needed. Works on Android (dominant in Kenya).	**Firebase Cloud Messaging (same — always free)** No upgrade needed. FCM remains free regardless of message volume.
WhatsApp Integration	**WhatsApp Business App (free, manual) → Meta Cloud API (free tier: 1,000 conversations/month)** Start with Meta's free Cloud API directly. 1,000 free service conversations/month. Enough for MVP parent comms.	**WhatsApp Business API via 360dialog (~$50/mo) for automation** Upgrade to a BSP (like 360dialog) when you need automated bulk messaging, webhooks, and message templates at scale.
Secrets Management	**Vercel/Railway built-in environment variables (free)** Use Vercel's built-in env var management for MVP. Simple, free, secure enough.	**Infisical Cloud Free → Doppler ($10/mo)** Add Infisical or Doppler when you have multiple environments (dev/staging/prod) and a team managing secrets.

7. Cost Summary: MVP vs. Production

Category	MVP Monthly Cost	Production Monthly Cost	Trigger to Upgrade
LLM API (Claude)	**~$10–30/mo (Haiku, heavy caching)**	**~$100–500/mo (Sonnet, 500+ learners)**	When lesson quality requires Sonnet, or revenue covers cost
Frontend Hosting	**$0 (Vercel Hobby)**	**$20/mo (Vercel Pro)**	Team growth or bandwidth >100GB/mo
Backend Services	**~$0–5/mo (Railway free credit)**	**$20–50/mo (Railway/Render)**	AI services exceed Railway free credit
Primary Database	**$0 (Supabase free)**	**$25/mo (Supabase Pro)**	DB size >400MB or need daily backups
Vector Database	**$0 (Weaviate sandbox)**	**$25–50/mo (Weaviate Cloud / self-hosted)**	Multiple grades ingested, >1GB chunks
Knowledge Graph	**$0 (Neo4j Community self-hosted)**	**$65/mo (Neo4j Aura) or $10/mo (Railway)**	Graph size or HA requirements
Storage (Portfolio)	**$0 (Supabase 1GB)**	**$0.015/GB (Cloudflare R2)**	Storage >1GB across families
Voice STT	**$0 (Whisper.cpp local)**	**$0.006/min (OpenAI API fallback)**	Local compute insufficient for traffic
Voice TTS	**$0 (Kokoro local)**	**$5–50/mo (ElevenLabs for premium tier)**	Premium subscription feature
Email	**$0 (Resend 3K/mo)**	**$20/mo (Resend Pro)**	>150 active families
WhatsApp	**$0 (Meta Cloud API 1K/mo)**	**$50/mo (BSP like 360dialog)**	Automated messaging at scale
Monitoring	**$0 (Sentry + PostHog free)**	**$26–50/mo (Sentry Team + PostHog)**	>5,000 errors/mo or >1M events/mo
CDN / Security	**$0 (Cloudflare free)**	**$20/mo (Cloudflare Pro)**	Paying users need advanced WAF
TOTAL ESTIMATE	**KES 0 – 4,000/mo ($0–30)**	**KES 50,000–100,000/mo ($380–770)**	When you have 200–500 paying families

KEY INSIGHT: At KES 500/month per family subscription, just 100 paying families generates KES 50,000/mo — which covers the full production infrastructure cost. Your upgrade path is self-funding.

8. Migration Roadmap: When to Upgrade

Think of the upgrade path in three stages tied to your learner and revenue milestones:

Stage 1: MVP (Weeks 1–16) — Target: 0–50 Families, KES 0 Spend

Everything runs on free tiers. Your only variable cost is the Anthropic API. Control this by:

- Generating lessons once and caching them in Supabase — never regenerate a lesson unless the curriculum changes
- Using Claude Haiku (cheapest model) for all generation during this phase
- Limiting voice STT to Whisper.cpp running locally on your Railway service (no per-call cost)
- Keeping the beta group small and intentional: 20–50 families, Grade 4 English only

Stage 2: Early Revenue (Weeks 17–36) — Target: 50–200 Families, KES 500/family/mo

You now have KES 25,000–100,000/month in revenue. Start upgrading the components that are causing friction:

- Supabase Free → Pro ($25/mo): triggered by database growth or need for automated backups
- Weaviate Sandbox → Cloud Starter ($25/mo): triggered by ingesting Grades 1–6 curriculum
- Vercel Hobby → Pro ($20/mo): triggered by team size or bandwidth
- Resend Free → Pro ($20/mo): triggered by exceeding 3,000 emails/month
- WhatsApp Meta free → 360dialog ($50/mo): triggered by needing automated bulk messaging

Estimated infrastructure cost at this stage: KES 15,000–25,000/month ($115–190). Well within revenue.

Stage 3: Scale (Weeks 37+) — Target: 200–1,000 Families

You are now a running business with predictable revenue. Upgrade the remaining high-impact components:

- Claude Haiku → Claude Sonnet: triggered by lesson quality feedback from families, budget permitting
- FastAPI on Railway → containerised microservices (Railway Team or AWS ECS): triggered by AI service performance
- In-memory cache → Upstash Redis ($10/mo): triggered by multiple server instances or distributed session needs
- Cloudflare Free → Pro ($20/mo): triggered by security requirements for paying users
- PostHog + Grafana for deep learning analytics: triggered by need to understand learner behaviour at scale
- Neo4j self-hosted → Neo4j Aura Professional: triggered by graph size, HA requirements, or team access needs

Estimated infrastructure cost at Stage 3: KES 50,000–100,000/month. At 500 families × KES 500, revenue is KES 250,000/month.

NEVER MIGRATE EVERYTHING AT ONCEThe most important rule: upgrade one component at a time, triggered by a specific measurable condition. This means you always know exactly why you are spending more, and you can validate that the upgrade solved the problem. A staggered upgrade approach also means zero downtime migrations — each service is independent.

9. Quick Reference: Free vs. Paid at a Glance

Component	MVP (Free)	Upgrade Trigger
Frontend	**Vercel Hobby (free)**	Team grows or >100GB bandwidth
Database	**Supabase Free (500MB)**	>400MB data or need daily backups
Auth	**NextAuth.js (always free)**	Never — no upgrade needed
Vector DB	**Weaviate Cloud Sandbox (1GB)**	Multiple grades ingested
Knowledge Graph	**Neo4j Community (self-hosted)**	HA required or graph >2GB
LLM	**Claude Haiku (API, low-cost)**	Lesson quality demands Sonnet
LLM Caching	**Supabase + in-memory (free)**	Upstash Redis at multi-instance scale
Embeddings	**sentence-transformers (local, free)**	Latency demands OpenAI embeddings
STT (Voice)	**Whisper.cpp (local, free)**	Device compute too low for local
TTS (Voice)	**Kokoro-82M (local, free)**	Premium voice tier for paid users
CI/CD	**GitHub Actions (2K min/mo free)**	Large team or >2K CI minutes
Error Monitoring	**Sentry Free (5K errors/mo)**	>5,000 errors/month
Analytics	**PostHog Cloud (1M events free)**	>1M events/month
CDN / DDoS	**Cloudflare Free**	Paying users needing advanced WAF
Email	**Resend Free (3K/mo)**	>150 active families
WhatsApp	**Meta Cloud API (1K conv/mo)**	Automated bulk messaging needed
Push Notifications	**Firebase FCM (always free)**	Never — no upgrade needed
Secrets	**Vercel env vars (free)**	Multi-environment team workflow
Portfolio Storage	**Supabase Storage (1GB)**	>1GB → Cloudflare R2
Background Jobs	**Vercel Cron + in-process (free)**	Reliable queuing needed at scale

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